

Dr. Babasaheb Ambedkar Technological University, Lonere

Supplementary Examination Summer 2024

Course : B. Tech.

Semester: V

Branch:Electrical Engineering / Electrical & Power / Electrical & Electronics / Electronics and Power

Subject Code : BTEEC503

Subject Name: Power Electronics

Duration : 03 Hours

Date: 05/07/2024

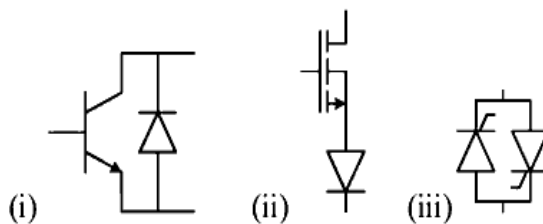
Maximum Marks : 60

INSTRUCTIONS TO CANDIDATES

1. All questions are compulsory.
2. Assume suitable data wherever necessary and state the assumptions made.
3. Diagrams / sketches should be given wherever necessary.
4. Use of logarithmic table, drawing instruments and non-programmable calculators is permitted.
5. Figures to the right indicate full marks.

Q. 01 Attempt any two the following.

- (a) Draw V-I characteristics with comments for the following combined devices in (i, ii & iii) when forward and reverse voltage is applied in presence of required control signal.



[CO,BL] (Marks)

[1,1] (06)

- (b) Draw neat circuit symbols and device characteristics (with required label) for the following: (i) Power Transistor (ii) Power MOSFET (iii) IGBT

[1,1] (06)

- (c) List and describe in 1-2 sentence, various specifications of the power device(s) to be taken into consideration to select for applications

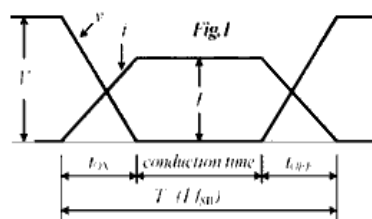
[1,1] (06)

Q. 02 Attempt any two of the following.

- (a) List generalized requirement-characteristics of gate / base drive circuits. Why isolation between control signal generating source and power circuit is essential?

[2,2] (06)

- (b) For the switching characteristics shown in fig. 1, assume linear variation for V & I with respect to time ' t ' during ' t_{ON} ' & ' t_{OFF} '. The OFF-state voltage is ' V ' and ON-state current is ' I ' with switching frequency of ' f_{SW} '. Assume the switch is ideal with regards to its static performance. Obtain the expression for average switching power loss.



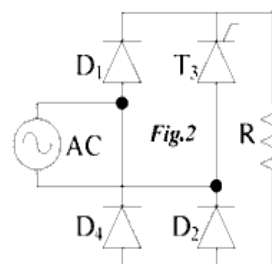
[2,3] (06)

- (c) With neat circuit diagram, waveform(s), and equation(s), describe UJT relaxation oscillator circuit useful as thyristor triggering circuit.

[2,2] (06)

Q. 03 Attempt any two of the following.

- (a) Draw all the input and out waveforms for the circuit shown in fig. 2 assuming the control pulse applied as per requirements.



[3,3] (06)

- (b) With necessary diagrams, waveforms, and equations, describe the effect of source impedance on the performance of 1-ph rectifiers. [3,2] (06)
- (c) Draw a neat circuit diagram and briefly describe the operation of Buck converter. Draw all the waveforms for continuous-current conduction mode of operation. [3,2] (06)

Q. 04 Attempt any two of the following.

- (a) A Buck converter has output voltage $V_0=5V$ when operated with 40% duty cycle at $f_{sw}=20$ kHz. The output current $I_0=200mA$, $L=1.0mH$ and $C=470\mu F$. Calculate ripples in the output voltage ΔV_0 (p-p) assuming all the components to be ideal. [3,4] (06)
- (b) A 1-ph half-wave controlled rectifier feeds 15Ω resistive-load. The source voltage is $230V$. Firing angle of the system is $\alpha=90^\circ$. Calculate, V_{dc} , I_{dc} , V_{rms} , I_{rms} , P_{dc} , P_{ac} , efficiency of the circuit, transformer utilization factor (TUF), VI -rating of the transformer, PIV of the power devices. [3,4] (06)
- (c) Give complete classification of inverters. [4,1] (06)

Q. 05 Attempt the following.

- (a) Give complete classification cycloconverters and write their applications. [4,1] (06)
- (b) Draw a neat diagram of 1-ph full-bridge (H-bridge) inverter. Explain its operation with all necessary waveforms for R-load condition. Write the equation for out-put voltage of this inverter. [4,2] (06)
